Chapter 6. Extension Fields

Extension fields, vector spaces, algebraic extensions, geometric constructions, finite fields

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References of the form "Theorem 27.25" without a chapter prefix refer to Fraleigh's numbering, used in brackets throughout both volumes.

6.1 Introduction to extension fields (§29)

Motivation. The polynomial x^2+1 has no root in $\mathbb{R}$, so we enlarge $\mathbb{R}$ to $\mathbb{C}$. Kronecker's theorem says this can always be done: every nonconstant polynomial over any field has a root in some larger field, namely $F[x]/\langle p(x)\rangle$. The rest of the chapter studies such extensions $E \supseteq F$, and the central question of Part 9, which polynomials are solvable by radicals, is phrased in their language.

A field E is an **extension field** of F if $F \leq E$ (a subfield).

Theorem 6.1 (29.3, Kronecker)

If F is a field and $f(x) \in F[x]$ is nonconstant, there is an extension field E of F and $\alpha \in E$ with $f(\alpha) = 0$.

Proof. Let $p(x)$ be an irreducible factor of f (Theorem 23.20). $\langle p\rangle$ is maximal (Theorem 27.25), so $E = F[x]/\langle p\rangle$ is a field. $a \mapsto a + \langle p\rangle$ is a one-to-one homomorphism $F \to E$ (its kernel is an ideal of the field F not containing 1), and we identify F with its image. For $\alpha = x + \langle p\rangle$, $p(\alpha) = p(x) + \langle p\rangle = 0$, so $f(\alpha) = 0$. ■

$\alpha \in E$ is **algebraic** over F if $f(\alpha) = 0$ for some nonzero $f \in F[x]$, and **transcendental** otherwise.

Theorem 6.2 (29.12)

Let $\phi_\alpha : F[x] \to E$ be the evaluation homomorphism. α is transcendental over F iff ϕ_α is one-to-one, i.e. an isomorphism of $F[x]$ onto its image (an integral domain).

Proof. $\ker \phi_\alpha = \{f : f(\alpha) = 0\}$ is $\{0\}$ exactly when α is transcendental. ■

Theorem 6.3 (29.13)

If $\alpha \in E$ is algebraic over F, there is an irreducible $p(x) \in F[x]$ with $p(\alpha) = 0$, unique up to a constant factor; it has minimal degree among nonzero polynomials vanishing at α, and $f(\alpha) = 0$ implies $p \mid f$.

Proof. $\ker \phi_\alpha$ is a nonzero ideal of $F[x]$, so equals $\langle p\rangle$ (Theorem 27.24) with p of minimal degree in it. p is irreducible: if $p = gh$ with lower degrees, then $g(\alpha)h(\alpha) = 0$ gives a nonzero polynomial of smaller degree vanishing at α. Every f with $f(\alpha) = 0$ lies in $\langle p\rangle$. If q is another irreducible with $q(\alpha) = 0$, then $p \mid q$, so $q = cp$. ■

The monic such p is $\mathrm{irr}(\alpha, F)$, and its degree is $\deg(\alpha, F)$. $F(\alpha)$ denotes the smallest subfield of E containing F and α (a **simple extension**).

Theorem 6.4 (29.18)

If α is algebraic over F of degree n, every $\beta \in F(\alpha)$ is uniquely $\beta = b_0 + b_1\alpha + \cdots + b_{n-1}\alpha^{n-1}$ with $b_i \in F$.

Proof. $\phi_\alpha[F[x]] \cong F[x]/\langle p\rangle$ is a field (Theorem 27.25), containing F and α, so it equals $F(\alpha)$. Thus $\beta = f(\alpha)$ for some f; dividing, $f = pq + r$ with $\deg r < n$, and $\beta = r(\alpha)$. If two such expressions agree, their difference is a polynomial of degree $< n$ vanishing at α, hence zero. ■

Example 6.1

$\mathbb{R}[x]/\langle x^2+1\rangle \cong \mathbb{R}(i) = \mathbb{C}$. x^2+x+1 is irreducible over $\mathbb{Z}_2$, so $\mathbb{Z}_2(\alpha) = \{0, 1, \alpha, 1+\alpha\}$ with $\alpha^2 = \alpha + 1$ is a field of 4 elements. π and e are transcendental over $\mathbb{Q}$ (Lindemann, Hermite).

6.2 Vector spaces (§30)

Motivation. An extension field E of F is automatically a vector space over F, so its size can be measured by a dimension, the degree $[E : F]$. Linear algebra is reviewed here because degree counting is the main computational tool for the rest of the course.

A **vector space** over F is an abelian group V with a scalar multiplication $F \times V \to V$ satisfying $a(\alpha+\beta) = a\alpha + a\beta$, $(a+b)\alpha = a\alpha + b\alpha$, $a(b\alpha) = (ab)\alpha$, $1\alpha = \alpha$.

Theorem 6.5 (30.5)

In a vector space, $0\alpha = 0$, $a0 = 0$, and $(-a)\alpha = a(-\alpha) = -(a\alpha)$.

Proof. $0\alpha + 0\alpha = (0+0)\alpha = 0\alpha$; cancel. Likewise $a0 = 0$. $a\alpha + (-a)\alpha = 0\alpha = 0$ and $a\alpha + a(-\alpha) = a0 = 0$. ∎

Vectors **span** V if every vector is a linear combination of them; they are **linearly independent** if $\sum a_i\alpha_i = 0$ only for all $a_i = 0$; a **basis** is an independent spanning set. V is **finite-dimensional** if a finite set spans it.

Lemma 6.6 (30.16)

If α is a linear combination of the β_i and each β_i is a linear combination of the γ_j, then α is a linear combination of the γ_j.

Proof. $\alpha = \sum_i b_i \sum_j c_{ij}\gamma_j = \sum_j (\sum_i b_i c_{ij})\gamma_j$. ∎

Theorem 6.7 (30.17)

Every finite spanning set of a vector space contains a basis.

Proof. Delete from the list $\alpha_1, \ldots, \alpha_n$ every α_i that is a linear combination of the preceding ones. The remaining vectors still span (Lemma 6.6), and they are independent: in a nontrivial relation, the last vector with nonzero coefficient would be a combination of earlier ones. ∎

Corollary 6.8 (30.18)

A finite-dimensional vector space has a finite basis.

Theorem 6.9 (30.19)

In a finite-dimensional space, any independent set $\{\alpha_1, \ldots, \alpha_r\}$ can be enlarged to a basis; and if $\{\beta_1, \ldots, \beta_n\}$ is any basis, then $r \le n$.

Proof. Apply the deletion process of Theorem 6.7 to the list $\alpha_1, \ldots, \alpha_r, \beta_1, \ldots, \beta_n$: no α_i is deleted (independence), so the result is a basis containing the α_i. For $r \le n$ (exchange): $\alpha_1 = \sum b_j\beta_j$ with some $b_j \neq 0$, say b_1; then $\alpha_1, \beta_2, \ldots, \beta_n$ is again a basis. Repeating, after k steps

$\alpha_1, \dots, \alpha_k, \beta_{k+1}, \dots, \beta_n$ is a basis (at step $k+1$, α_{k+1} has a nonzero coefficient on some remaining β, by independence of the α's). If $r > n$, after n steps $\alpha_1, \dots, \alpha_n$ is a basis and α_{n+1} is a combination of them, a contradiction. ■

Corollary 6.10 (30.20)

Any two bases of a finite-dimensional vector space have the same number of elements, the **dimension**.

Theorem 6.11 (30.23)

If α is algebraic over F of degree n, then $F(\alpha)$ is an n-dimensional vector space over F with basis $1, \alpha, \dots, \alpha^{n-1}$, and every $\beta \in F(\alpha)$ is algebraic over F with $\deg(\beta, F) \le n$.

Proof. The basis statement is Theorem 6.4. The $n+1$ elements $1, \beta, \dots, \beta^n$ are dependent (Theorem 6.9), giving a nonzero polynomial of degree $\le n$ with zero β. ■

6.3 Algebraic extensions (§31)

Motivation. Degrees multiply in towers, $[K : F] = [K : E][E : F]$. This innocent formula is surprisingly powerful: it shows that algebraic numbers form a field and, in the next section, settles classical construction problems. We also construct the algebraic closure, a field in which every polynomial splits, and prove the fundamental theorem of algebra for $\mathbb{C}$.

E is an **algebraic extension** of F if every element of E is algebraic over F, and a **finite extension** of degree $[E : F]$ if it has finite dimension $[E : F]$ as a vector space over F.

Theorem 6.12 (31.3)

A finite extension is algebraic.

Proof. If $[E : F] = n$, then $1, \alpha, \dots, \alpha^n$ are dependent for every $\alpha \in E$. ■

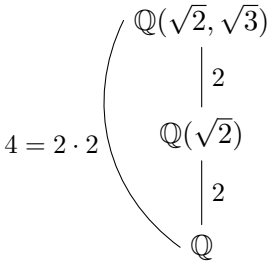


Figure. *Degrees multiply in a tower:* $[\mathbb{Q}(\sqrt{2}, \sqrt{3}) : \mathbb{Q}] = [\mathbb{Q}(\sqrt{2}, \sqrt{3}) : \mathbb{Q}(\sqrt{2})]\,[\mathbb{Q}(\sqrt{2}) : \mathbb{Q}] = 2 \cdot 2$.

Theorem 6.13 (31.4)

If E is a finite extension of F and K a finite extension of E, then K is a finite extension of F and $[K : F] = [K : E][E : F]$.

Proof. Let $\alpha_1, \dots, \alpha_n$ be a basis of E over F and $\beta_1, \dots, \beta_m$ a basis of K over E. Then $\{\alpha_i \beta_j\}$ spans K over F: $\gamma = \sum_j b_j \beta_j = \sum_j \sum_i a_{ij} \alpha_i \beta_j$. It is independent: if $\sum_{i,j} c_{ij} \alpha_i \beta_j = 0$, then $\sum_j (\sum_i c_{ij} \alpha_i) \beta_j = 0$, so each $\sum_i c_{ij} \alpha_i = 0$ and each $c_{ij} = 0$. ■

Corollary 6.14 (31.6)

For a tower $F_1 \leq F_2 \leq \cdots \leq F_r$ of finite extensions, $[F_r : F_1] = [F_r : F_{r-1}] \cdots [F_2 : F_1]$.

Corollary 6.15 (31.7)

If α is algebraic over F and $\beta \in F(\alpha)$, then $\deg(\beta, F)$ divides $\deg(\alpha, F)$.

Proof. $[F(\alpha) : F] = [F(\alpha) : F(\beta)][F(\beta) : F]$ and $[F(\beta) : F] = \deg(\beta, F)$. ■

So $\mathbb{Q}(\sqrt{2})$ contains no $\sqrt[3]{2}$ $(3 \nmid 2)$. $F(\alpha_1, \ldots, \alpha_n)$ is the smallest subfield containing F and the α_i.

Theorem 6.16 (31.11)

An algebraic extension E of F is finitely generated, $E = F(\alpha_1, \ldots, \alpha_n)$, iff it is a finite extension.

Proof. If $E = F(\alpha_1, \ldots, \alpha_n)$, each α_i is algebraic over $F(\alpha_1, \ldots, \alpha_{i-1})$, so each step is finite and $[E : F] < \infty$ (Corollary 6.14). Conversely, a basis over F generates E. ■

Theorem 6.17 (31.12)

For an extension E of F, $\overline{F}_E = \{\alpha \in E : \alpha \text{ algebraic over } F\}$ is a subfield of E (the **algebraic closure of F in E**).

Proof. If $\alpha, \beta \in \overline{F}_E$, then $F(\alpha, \beta)$ is a finite extension (Theorem 6.16), hence algebraic (Theorem 6.12), and contains $\alpha \pm \beta$, $\alpha\beta$ and α/β. ■

Corollary 6.18 (31.13)

The algebraic numbers (complex numbers algebraic over $\mathbb{Q}$) form a field.

F is **algebraically closed** if every nonconstant $f \in F[x]$ has a zero in F; an **algebraic closure** $\overline{F}$ of F is an algebraic extension that is algebraically closed.

Theorem 6.19 (31.15)

F is algebraically closed iff every nonconstant polynomial in $F[x]$ factors into linear factors in $F[x]$.

Proof. If F is algebraically closed, a zero a gives $f = (x - a)g$ (factor theorem); induct on degree. The converse is clear. ■

Corollary 6.20 (31.16)

An algebraically closed field has no proper algebraic extension.

Proof. If α is algebraic over F, $\text{irr}(\alpha, F)$ factors into linear factors over F, so is linear and $\alpha \in F$. ■

Lemma 6.21 (31.21, Zorn's lemma)

If every chain in a nonempty partially ordered set S has an upper bound in S, then S has a maximal element.

Proof. Zorn's lemma is equivalent to the axiom of choice and is taken as an axiom of set theory (see the Topology notes for its equivalence with the well-ordering theorem). ■

Theorem 6.22 (31.17 and 31.22)

Every field F has an algebraic closure $\overline{F}$.

Proof [Supplement]. (Artin.)

Step 1. An extension in which every nonconstant $f \in F[x]$ has a zero. Introduce an indeterminate x_f for each nonconstant $f \in F[x]$, let $R = F[\{x_f\}]$ and I the ideal generated by all $f(x_f)$. $I \neq R$: otherwise $1 = \sum_{i \le k} g_i f_i(x_{f_i})$; by Kronecker's theorem (Theorem 6.1, repeatedly) there is an extension in which each f_i has a zero α_i, and substituting $x_{f_i} = \alpha_i$ (other variables 0) gives $1 = 0$. By Zorn's lemma, the proper ideals containing I (ordered by inclusion; the union of a chain is proper since it misses 1) have a maximal element M. $E_1 = R/M$ is a field containing F, in which $x_f + M$ is a zero of f; and E_1 is generated by these algebraic elements, so it is algebraic over F (Theorem 6.17).

Step 2. Repeat to get $F \le E_1 \le E_2 \le \cdots$, each algebraic over the previous one, and $L = \bigcup E_n$, a field algebraic over F (algebraic over algebraic is algebraic: if α is a zero of $\sum c_i x^i$ with c_i algebraic over F, then $F(c_0, \ldots, c_m, \alpha)$ is finite over F). Each nonconstant $h \in F[x]$ splits into linear factors over L: it has a zero in E_1, the quotient (with coefficients in E_1) has a zero in E_2, and so on. Now let $g \in L[x]$ be nonconstant, and α a zero of g in some extension K of L (Kronecker). α is algebraic over L, hence over F; let $h = \operatorname{irr}(\alpha, F) = \prod(x - \beta_i)$ with $\beta_i \in L$. As $h(\alpha) = 0$ in K, $\alpha = \beta_i \in L$ for some i. So g has a zero in L, and L is an algebraic closure of F. ■

Theorem 6.23 (31.18, fundamental theorem of algebra)

$\mathbb{C}$ is algebraically closed.

Proof [Supplement]. Let $f \in \mathbb{C}[x]$ be nonconstant of degree n. Since $|f(z)| \to \infty$ as $|z| \to \infty$, $|f|$ attains a minimum at some z_0 (on a large closed disc, by compactness). Suppose $f(z_0) \neq 0$. Expanding, $f(z_0 + w) = f(z_0) + cw^k + w^{k+1}h(w)$ with $c \neq 0$, $k \ge 1$, h a polynomial. Choose $\omega \in \mathbb{C}$ with $\omega^k = -f(z_0)/c$ (k-th roots exist in $\mathbb{C}$). For $0 < t < 1$, $f(z_0 + t\omega) = f(z_0)(1 - t^k) + t^{k+1}\omega^{k+1}h(t\omega)$, so $|f(z_0 + t\omega)| \le |f(z_0)|(1 - t^k) + Mt^{k+1} < |f(z_0)|$ for small t, where M bounds $|\omega^{k+1}h(t\omega)|$; a contradiction. So $f(z_0) = 0$. ■

A purely algebraic proof via Galois theory and the Sylow theorems is given after Theorem 53.6.

6.4 Geometric constructions (§32)

Motivation. Can one trisect an angle or double a cube with ruler and compass? Constructible numbers lie in towers of quadratic extensions, so their degrees are powers of 2; since $\cos 20^\circ$ and $\sqrt[3]{2}$ have degree 3, both constructions are impossible. Two thousand years of attempts are settled by the tower formula.

A real number α is **constructible** if a segment of length $|\alpha|$ can be constructed with straightedge and compass from a given unit segment.

Theorem 6.24 (32.1)

If α, β are constructible, so are $\alpha \pm \beta$, $\alpha\beta$ and α/β ($\beta \neq 0$).

Proof. Sums and differences: lay off segments on a line. Products and quotients: by similar triangles (on two rays from a point mark 1 and β on one, α on the other; the parallel through β to the line joining 1 and α cuts off $\alpha\beta$, and similarly for α/β). ■

Corollary 6.25 (32.5)

The constructible real numbers form a subfield $\mathcal{F}$ of $\mathbb{R}$ containing $\mathbb{Q}$.

Square roots of positive constructible numbers are constructible (on a segment of length $1+\alpha$ draw a semicircle; the perpendicular at the junction has length $\sqrt{\alpha}$).

Theorem 6.26 (32.6)

$\mathcal{F}$ consists exactly of the real numbers obtainable from $\mathbb{Q}$ by finitely many field operations and square roots of positive numbers.

Proof. Such numbers are constructible by the above. Conversely, all points constructed from points with coordinates in a subfield F of $\mathbb{R}$ lie in F or in $F(\sqrt{a})$, $a \in F$, $a > 0$: intersecting two lines with equations over F gives coordinates in F; intersecting a line and a circle, or two circles (subtracting the equations reduces to a line and a circle), leads to a quadratic equation over F. ■

Corollary 6.27 (32.8)

If γ is constructible and $\gamma \notin \mathbb{Q}$, there are $\gamma_1, \ldots, \gamma_n = \gamma$ with $[\mathbb{Q}(\gamma_1, \ldots, \gamma_i) : \mathbb{Q}(\gamma_1, \ldots, \gamma_{i-1})] = 2$. In particular $[\mathbb{Q}(\gamma) : \mathbb{Q}] = 2^r$.

Proof. By Theorem 6.26 and Corollary 6.15, as $\gamma \in \mathbb{Q}(\gamma_1, \ldots, \gamma_n)$ of degree 2^n. ■

Theorem 6.28 (32.9)

Doubling the cube is impossible.

Proof. It requires constructing $\sqrt[3]{2}$, of degree 3 over $\mathbb{Q}$ ($x^3 - 2$ is irreducible by Eisenstein), not a power of 2. ■

Theorem 6.29 (32.10)

Squaring the circle is impossible.

Proof. For the unit circle it requires $\sqrt{\pi}$, hence π, to be constructible, hence algebraic; but π is transcendental (Lindemann). ■

Theorem 6.30 (32.11)

There is an angle that cannot be trisected; e.g. $60°$.

Proof. Trisecting $60°$ means constructing $\alpha = \cos 20°$. From $\cos 3\theta = 4\cos^3\theta - 3\cos\theta$, $8\alpha^3 - 6\alpha - 1 = 0$, i.e. 2α is a zero of $x^3 - 3x - 1$, which has no rational zero (± 1 fail), so is irreducible of degree 3. ■

6.5 Finite fields (§33)

Motivation. Finite fields appear in coding theory, cryptography and number theory. They turn out to be completely classified: for every prime power p^n there is exactly one field of that order up to isomorphism, it is the splitting field of $x^{p^n} - x$, and its multiplicative group is cyclic.

Theorem 6.31 (33.1)

If E is an extension of degree n of a finite field F with q elements, then $|E| = q^n$.

Proof. Each element is uniquely $\sum_{i \le n} b_i \alpha_i$ for a basis α_i. ■

Corollary 6.32 (33.2)

A finite field of characteristic p has p^n elements for some n.

Proof. It is a finite extension of its prime field $\mathbb{Z}_p$. ■

Theorem 6.33 (33.3)

If $E \le \overline{\mathbb{Z}_p}$ has p^n elements, its elements are exactly the zeros of $x^{p^n} - x$ in $\overline{\mathbb{Z}_p}$.

Proof. E^* has order $p^n - 1$, so $\alpha^{p^n - 1} = 1$ for $\alpha \neq 0$ and $\alpha^{p^n} = \alpha$ for all $\alpha \in E$. These are p^n zeros, all the zeros of the polynomial of degree p^n. ■

α is a **primitive n-th root of unity** if $\alpha^n = 1$ and $\alpha^m \neq 1$ for $0 < m < n$.

Theorem 6.34 (33.5)

The multiplicative group of a finite field is cyclic.

Proof. Corollary 23.6. ■

Corollary 6.35 (33.6)

A finite extension of a finite field is simple.

Proof. $E = F(\alpha)$ for a generator α of E^*. ■

Lemma 6.36 (33.8)

If F has characteristic p, $x^{p^n} - x$ has p^n distinct zeros in $\overline{F}$.

Proof. It splits in $\overline{F}$; a repeated zero α would be a zero of the derivative $p^n x^{p^n - 1} - 1 = -1$, which has no zeros. (Directly: if $f = (x - \alpha)^2 g$, then $f' = (x - \alpha)[2g + (x - \alpha)g']$ vanishes at α.) ■

Lemma 6.37 (33.9)

In characteristic p, $(\alpha + \beta)^{p^n} = \alpha^{p^n} + \beta^{p^n}$.

Proof. For $n = 1$, $p \mid \binom{p}{k}$ for $0 < k < p$; induct. ■

Theorem 6.38 (33.10)

For every prime power p^n there is a finite field $\mathrm{GF}(p^n)$ of p^n elements.

Proof. The zeros of $x^{p^n} - x$ in $\overline{\mathbb{Z}_p}$ (p^n of them, Lemma 6.36) form a subfield: they are closed under sums (Lemma 6.37), products, negatives and inverses ($(\alpha^{-1})^{p^n} = \alpha^{-1}$). ■

Corollary 6.39 (33.11)

If F is finite, then for every $n \geq 1$ there is an irreducible polynomial of degree n in $F[x]$.

Proof. Let $|F| = p^r$. The zeros of $x^{p^{rn}} - x$ in $\overline{F}$ form a field $E \supseteq F$ (the elements of F satisfy $\alpha^{p^r} = \alpha$) with $[E : F] = n$ (Theorem 6.31). $E = F(\beta)$ by Corollary 6.35, and $\mathrm{irr}(\beta, F)$ has degree n. ■

Theorem 6.40 (33.12)

Any two fields with p^n elements are isomorphic.

Proof. Both are splitting fields of $x^{p^n} - x$ over $\mathbb{Z}_p$ (Theorem 6.33), hence isomorphic (Corollary 49.5 and the uniqueness of splitting fields, Section 9.3). Directly: let $E = \mathbb{Z}_p(\alpha)$ and $f = \mathrm{irr}(\alpha, \mathbb{Z}_p)$; f divides $x^{p^n} - x$, which splits in E', so f has a zero $\alpha' \in E'$, and $\mathbb{Z}_p(\alpha) \cong \mathbb{Z}_p[x]/\langle f \rangle \cong \mathbb{Z}_p(\alpha') \leq E'$; comparing orders, $\mathbb{Z}_p(\alpha') = E'$. ■